Cross-Platform Sentiment Classification: Analyzing Domain Shift Between Reddit and Twitter Using Machine Learning

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***Abstract*—Social media platforms differ substantially in their linguistic style, text length, and expressive patterns, posing a significant challenge for sentiment classification models trained on one platform and deployed on another. This paper investigates cross-domain sentiment generalization by training multiple classical machine learning models on Reddit data and evaluating them on Twitter data, both annotated with three-class sentiment labels: Negative (–1), Neutral (0), and Positive (1). Four baseline classifiers are compared—Logistic Regression, Linear Support Vector Machine (SVM), Naive Bayes, and Random Forest—using TF-IDF features with unigrams and bigrams over a vocabulary of 5,000 terms. Each platform provides 3,000 balanced samples (1,000 per class). The best-performing model, Linear SVM, achieves a weighted F1 score of 0.720 on the Reddit validation set, dropping to 0.653 on the Twitter test set, representing a degradation of 9.3%. We analyze the causes of this degradation through linguistic compression analysis (Pearson $r = 0.12$ between Type-Token Ratio compression and F1 drop) and model calibration curves (Expected Calibration Error of 0.13 for the Negative class). CORAL (CORrelation ALignment) domain adaptation was applied but produced a negative improvement of -0.047 in weighted F1, confirming that feature-level alignment is insufficient for cross-platform text classification. The Negative sentiment class poses the highest real-world risk, with a missed case rate of 6.96% in a mental health monitoring context. This work belongs to Project Type 2: *Understanding Existing Work and Comparing it with Different Datasets*.**

***Index Terms*—sentiment analysis, cross-platform classification, domain shift, TF-IDF, CORAL, Reddit, Twitter, machine learning**

I. Introduction

A. Problem Statement

Sentiment analysis is a core task in natural language processing (NLP) that involves classifying textual content by its emotional polarity—typically positive, negative, or neutral. While sentiment models achieve high accuracy on their training domain, they frequently fail to generalize when deployed in a different context or platform. This phenomenon is known as *domain shift* or *dataset shift*.

Reddit and Twitter are two of the most widely used social media platforms for opinion mining research, yet they differ significantly in text characteristics. Reddit posts tend to be longer, more structured, and richer in vocabulary (averaging 28.6 words per post), while Twitter posts are constrained to short, highly informal content (averaging 20.3 words per post). This structural difference of 8.3 words on average creates measurable distribution mismatch between the two platforms. A model trained on Reddit is unlikely to perform equally well on Twitter without adaptation.

B. Motivation and Objectives

The motivation for this work stems from the practical need to deploy sentiment classifiers that can generalize across social media platforms without requiring re-labeling of data. Such capability is essential for mental health monitoring, brand sentiment tracking, public opinion analysis, and crisis detection.

The objectives of this project are:

- Train and compare multiple classical ML classifiers for three-class sentiment classification using Reddit as the training domain.
- Evaluate model generalization on Twitter data as an unseen test domain.
- Quantify and explain the performance drop through linguistic analysis.
- Analyze model calibration on the cross-domain test set.
- Investigate whether CORAL domain adaptation can reduce the generalization gap.
- Assess the real-world risk of cross-domain failure, particularly for negative sentiment detection in mental health contexts.

Project Type: This project belongs to Type 2 — *Understanding Existing Work and Comparing it with Different Datasets*. The project reproduces and compares established ML classifiers and analyzes their cross-platform generalization behavior.

II. Background Study / Related Work

A. Sentiment Analysis in Social Media

Sentiment analysis on social media has been well-studied since the mid-2000s. Early approaches relied on lexicon-based methods such as VADER [1] and SentiWordNet. Supervised machine learning methods using TF-IDF features with classifiers such as Support Vector Machines and Naive Bayes became standard baselines, demonstrating competitive performance on in-domain benchmarks [2].

B. Domain Shift in NLP

Domain adaptation in NLP addresses the challenge of applying models trained on one data distribution (source domain) to a different distribution (target domain). Ben-David et al. [3] formalized the theoretical framework for domain adaptation, showing that performance in the target domain is bounded by the source domain performance and the divergence between distributions. Social media platforms represent a particularly challenging case of domain shift due to differences in register, vocabulary, and linguistic style.

C. Cross-Platform Challenges

Studies have shown significant accuracy drops when transferring sentiment classifiers between social media platforms [4]. Twitter introduces unique challenges including hashtags, mentions, slang, and character constraints. Reddit, by contrast, features longer discourse with threaded discussions. These structural differences create both lexical and syntactic distribution shift, which this work quantifies empirically.

D. CORAL Domain Adaptation

Sun et al. [5] proposed CORAL (CORrelation ALignment), a feature-level domain adaptation method that aligns the second-order statistics (covariance matrices) of source and target feature distributions. While effective for image classification, its applicability to high-dimensional sparse text features remains limited [6], as confirmed by our experimental results.

E. Limitations of Existing Work

Most prior work focuses on binary sentiment or fine-grained emotion classification rather than three-class sentiment in a cross-platform setting. Moreover, few studies provide a systematic analysis of *why* performance drops—i.e., which linguistic features contribute most to the degradation. This paper addresses both gaps.

III. Proposed Approach / Methodology

A. System Design Overview

The system is designed as a cross-platform sentiment classification pipeline with the following stages: data loading and preprocessing, feature engineering, model training and evaluation, diagnostic analysis (linguistic compression and calibration), and domain adaptation via CORAL. Fig. **??** illustrates the overall workflow.

B. Workflow Description

The pipeline proceeds through the following phases:

1) **Data Loading and Label Alignment:** Both Reddit and Twitter datasets are loaded, column names are standardized, and only common sentiment labels $\{-1, 0, 1\}$ across both platforms are retained. Reddit yields 37,249 samples and Twitter 162,973 samples before balancing.
2) **Text Cleaning:** Raw text is lowercased; URLs, mentions, and hashtags (converted to plain words) are removed; very short texts (under 10 characters) are filtered.
3) **Class Balancing:** Both datasets are balanced by undersampling to a maximum of 1,000 samples per class, yielding 3,000 samples per platform.
4) **Feature Engineering:** TF-IDF with unigram and bigram features (vocabulary size: 5,000) is applied. The vectorizer is fit only on Reddit training data to prevent data leakage.
5) **Model Training:** Four classifiers are trained on Reddit (2,400 training samples) and evaluated on Reddit validation (600 samples) and Twitter test (3,000 samples).
6) **Diagnostic Analysis:** Linguistic features are extracted and correlated with F1 performance drop. Calibration curves detect model overconfidence.
7) **Domain Adaptation:** CORAL is applied to align source and target feature covariances using a reduced vocabulary of 3,000 features.
8) **Risk Assessment:** Misclassification rates for negative sentiment are analyzed from a mental health monitoring perspective.

C. Algorithms Used

Logistic Regression with L2 regularization and balanced class weights, trained with a maximum of 1,000 iterations. Supports probability outputs used in calibration analysis.

Linear SVM (LinearSVC) with balanced class weights and a maximum of 2,000 iterations. Known to perform well on high-dimensional sparse TF-IDF representations and achieves the best cross-domain performance in our experiments.

Multinomial Naive Bayes with Laplace smoothing ($\alpha = 0.1$). Provides a fast probabilistic baseline.

Random Forest with 100 decision trees, balanced class weights, and parallel training. Captures non-linear feature interactions but is slower (2.3 seconds training time).

IV. Datasets and Models Used

A. Dataset Description

Two publicly available datasets from Kaggle were used.

Reddit Dataset: Comments from various subreddits labeled with sentiment categories (-1: Negative, 0: Neutral, 1: Positive). Raw dataset contains 37,249 samples after label alignment. Used exclusively for training and validation.

Twitter Dataset: Tweets labeled with the same three sentiment classes. Raw dataset contains 162,973 samples. Reserved entirely as the unseen cross-domain test set. After filtering and balancing, 3,000 samples are retained.

B. Data Statistics

Table I summarizes dataset sizes after preprocessing and balancing.

TABLE I
DATASET STATISTICS AFTER PREPROCESSING AND BALANCING

Split	Platform	Classes	Samples/Class
Train (80%)	Reddit	3	800 (2,400 total)
Validation (20%)	Reddit	3	200 (600 total)
Test	Twitter	3	1,000 (3,000 total)

C. Text Length Analysis

A key observation is the significant difference in average text length between the two platforms. Reddit posts average 28.6 words while Twitter posts average 20.3 words, a difference of 8.3 words. The full distribution shown in Fig. 1 reveals that Reddit has a much longer tail, with some posts exceeding 1,000 words, while Twitter is tightly concentrated at short lengths.

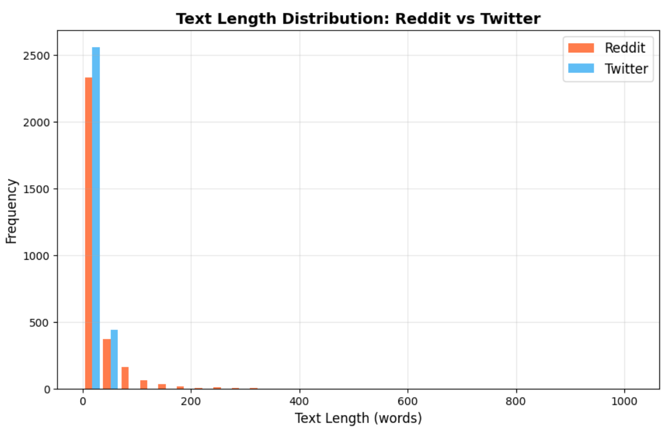

Fig. 1. Text length distribution for Reddit (orange) and Twitter (blue). Reddit texts are significantly longer with a heavier tail; Twitter texts are concentrated at shorter lengths.

D. Data Preprocessing Steps

The following cleaning steps were applied to both datasets:

- Lowercasing all text.
- Removing URLs (`http://`, `https://`, `www.`).
- Removing user mentions (`@username`).
- Converting hashtags (`#word`) to plain words.
- Stripping extra whitespace.
- Removing texts shorter than 10 characters.

E. TF-IDF Feature Engineering

TF-IDF (Term Frequency-Inverse Document Frequency) was configured with the following parameters:

- Maximum features: 5,000 (baseline) / 3,000 (CORAL experiment).
- N-gram range: (1, 2) — unigrams and bigrams.
- Minimum document frequency: 3.
- Maximum document frequency: 0.8 (removes very common words).
- Sublinear TF scaling enabled.

The resulting feature matrix shape was $(2400, 4971)$, indicating that 4,971 of the 5,000 requested features were populated by the Reddit vocabulary. The vectorizer was fitted exclusively on Reddit training data and applied (transformed only) to validation and test sets, preventing data leakage.

V. IMPLEMENTATION DETAILS

A. Tools and Libraries

- **Python 3.x** — primary programming language.
- **scikit-learn** [9] — classifiers, TF-IDF, metrics, calibration curves.
- **pandas / NumPy** — data manipulation and numerical computation.
- **matplotlib** — visualization of all figures.
- **TextBlob / VADER (NLTK)** [10] — linguistic feature extraction.
- **SciPy** — fractional matrix power for CORAL transformation.
- **tqdm** — progress tracking during feature extraction.
- **Google Colab** — cloud CPU/GPU execution environment (free tier).

B. Project Structure

The project is organized as follows:

- `/data/` — processed CSV files (`reddit_processed.csv`, `twitter_processed.csv`, linguistic feature files).
- `/models/` — serialized model files (`best_model.pkl`, `vectorizer.pkl`, `logistic_model.pkl`, `coral_model.pkl`).
- `/results/` — CSV result tables (`model_leaderboard.csv`, `per_emotion_performance.csv`, `calibration_summary.csv`, `coral_results.csv`, `clinical_risk_summary.csv`).
- `/visualisations/` — saved figures (`.png`).

C. Model Persistence

All trained models are serialized using Python's `pickle` library. The best-performing model (Linear SVM), the TF-IDF vectorizer, the CORAL-adapted Logistic Regression model, and the base Logistic Regression model (for calibration) are saved separately for reproducibility.

VI. RESULT ANALYSIS

A. Baseline Model Performance

Table II summarizes the performance of all four models on the Reddit validation set (in-domain) and the Twitter test set (cross-domain). Linear SVM achieves the highest cross-domain performance and is selected as the best model.

TABLE II
MODEL PERFORMANCE: REDDIT VALIDATION VS. TWITTER TEST

Model	RD Acc	RD F1	TW Acc	TW F1	Drop%
Linear SVM	0.720	0.720	0.653	0.653	9.3%
Logistic Reg.	0.712	0.712	0.628	0.628	11.8%
Random Forest	0.692	0.684	0.613	0.609	11.4%
Naive Bayes	0.602	0.597	0.543	0.537	9.8%

RD = Reddit, TW = Twitter. Sorted by Twitter F1 (descending).

All four models exhibit a consistent performance drop when evaluated on the unseen Twitter domain. Linear SVM incurs the smallest absolute accuracy drop (0.067), while Logistic Regression suffers the largest drop (11.8%). Naive Bayes, while the weakest overall, has a relatively small percentage drop (9.8%) due to its lower in-domain baseline.

B. Domain Shift Visualization

Fig. 2 shows the drop in weighted F1 score when moving from the Reddit validation set to the Twitter test set across all four models. A consistent decline is observed for all classifiers, confirming the presence of domain shift.

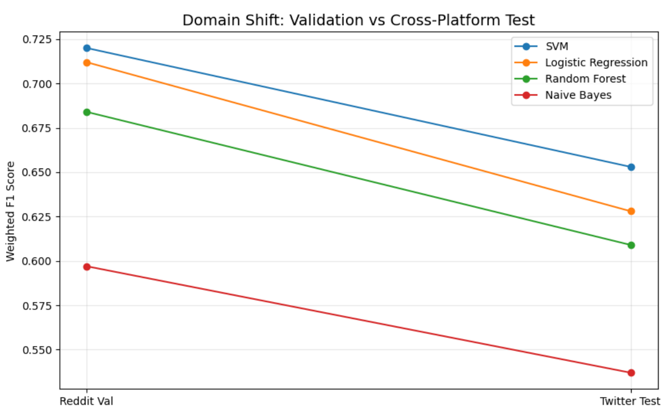

Fig. 2. Weighted F1 score on Reddit validation vs. Twitter test for all four models. All models exhibit a measurable and consistent performance drop, with SVM retaining the highest cross-domain score (0.653).

C. Confusion Matrix Analysis

The confusion matrix for the best-performing model (Linear SVM) on the Twitter test set is shown in Fig. 3. The matrix reveals that:

- The model correctly classifies 634 of 1,000 Negative samples (63.4% recall).
- The Neutral class achieves the highest correct prediction count (721 of 1,000, 72.1% recall).
- 216 Negative samples are misclassified as Neutral, and 150 as Positive.
- The Positive class achieves 60.5% recall (605 correct), with 210 misclassified as Neutral.

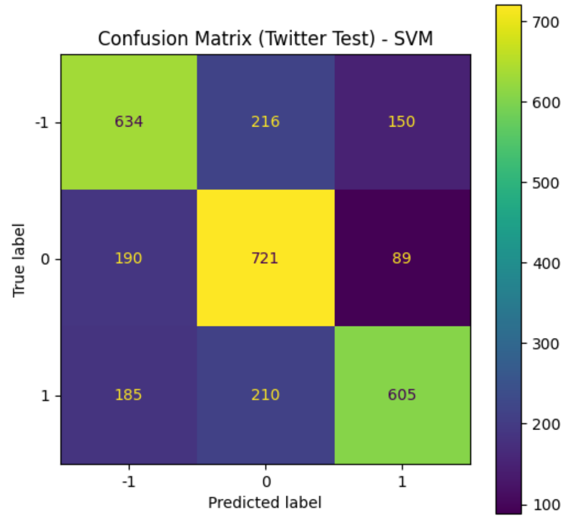

Fig. 3. Confusion matrix of Linear SVM (best model) on the Twitter test set. Rows represent true labels; columns represent predicted labels. Values in diagonal represent correct predictions.

TABLE III
PER-CLASS PERFORMANCE OF LINEAR SVM (REDDIT VALIDATION VS. TWITTER TEST)

Platform	Class	Precision	Recall	F1
Reddit Val	Negative (–1)	0.682	0.675	0.678
Reddit Val	Neutral (0)	0.748	0.755	0.751
Reddit Val	Positive (1)	0.730	0.730	0.730
Twitter Test	Negative (–1)	0.628	0.634	0.631
Twitter Test	Neutral (0)	0.630	0.721	0.672
Twitter Test	Positive (1)	0.717	0.605	0.656

D. Per-Class Classification Report

Table III presents the full per-class precision, recall, and F1 scores for the best model (SVM) on both platforms.

The Neutral class achieves the highest in-domain F1 (0.751) and maintains reasonable cross-domain performance (0.672). The Negative class shows a moderate drop from 0.678 to 0.631. The Positive class suffers particularly in recall (dropping from 0.730 to 0.605), indicating difficulty recognizing positive sentiment expressed in Twitter's informal style.

E. Error Analysis

Manual inspection of 20 randomly sampled misclassified Twitter examples revealed the following qualitative patterns:

- **Political sarcasm:** Posts appearing positive but labeled negative due to ironic framing (e.g., apparent praise for a politician masking criticism).
- **Informal and domain-specific vocabulary:** Slang, political references, and abbreviations not present in Reddit training data.
- **Ambiguous short texts:** Very short posts lacking sufficient context for reliable three-class classification.
- **Negation handling failures:** Cases where the model incorrectly weighs negative and positive terms (e.g.,

predicting Positive when True Label is Negative due to presence of positive trigger words).

A total of 1,040 out of 3,000 Twitter test samples were misclassified.

VII. Linguistic Compression Analysis

A. Extracted Linguistic Features

To understand the structural causes of performance degradation, ten linguistic features were extracted from both platforms per sentiment class:

- **Type-Token Ratio (TTR):** vocabulary richness (unique words / total words).
- Average word length.
- Exclamation mark density.
- Question mark density.
- Emoji count.
- VADER positive word ratio.
- VADER negative word ratio.
- TextBlob sentiment polarity.
- First-person pronoun density.
- Sentence count.

B. Platform-Level Differences

Table IV summarizes the key linguistic differences per sentiment class, computed as the difference (Reddit minus Twitter) in mean feature values.

TABLE IV
Linguistic Compression by Sentiment Class (Reddit − Twitter)

Class	Vocab. Change (TTR)	Sentence Structure	Sentiment Shift
Negative	−0.025	0.000	+0.006
Neutral	−0.005	0.000	0.000
Positive	−0.048	0.000	−0.023

Negative values indicate Twitter is richer in that feature.
Vocabulary Change = ΔTTR; Sentence Structure = Δsentence count.

The TTR differences are negative across all classes, indicating that Twitter vocabulary is actually more diverse relative to text length in these samples, though with shallower semantic depth. Reddit texts exhibit longer sentences (higher word count per post) and richer multi-sentence structure.

C. Correlation with F1 Drop

The scatter plot in Fig. 4 shows the relationship between vocabulary compression (TTR difference) and per-class F1 drop. The Pearson correlation coefficient is $r = 0.12$, confirming a weak-to-moderate positive relationship. The Positive class exhibits the largest TTR compression (-0.048) alongside a substantial F1 drop (0.074), while the Neutral class shows minimal TTR change (-0.005) yet also experiences the highest F1 drop (0.080), indicating that additional factors beyond vocabulary richness—such as semantic ambiguity and context dependence—also contribute to cross-domain failure.

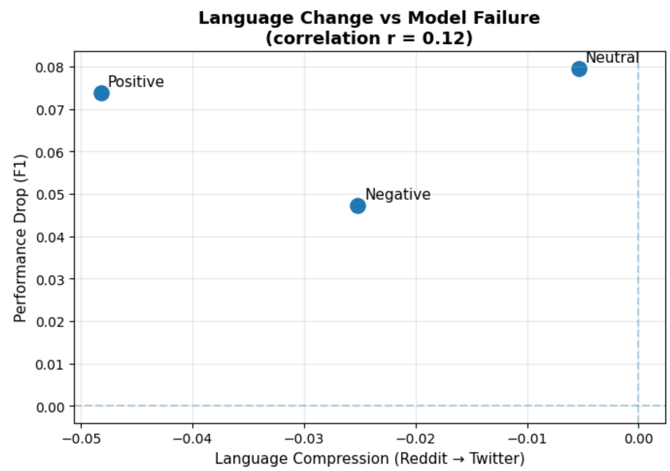

Fig. 4. Scatter plot of linguistic compression (TTR difference: Reddit − Twitter) versus per-class F1 performance drop. Pearson $r = 0.12$ suggests a weak positive correlation, indicating vocabulary compression is one contributing factor among several.

VIII. Calibration Analysis

Model calibration refers to the alignment between a model's predicted probability and the empirical accuracy on that subset of samples. A perfectly calibrated model should produce a predicted probability of 0.8 corresponding to an actual accuracy of 0.8 on the corresponding samples.

The calibration curves in Fig. 5 show that the Logistic Regression model trained on Reddit is *overconfident* when tested on Twitter data. All three class curves deviate above the perfect calibration diagonal, especially at higher predicted probabilities, indicating the model assigns high confidence to many predictions that are actually wrong.

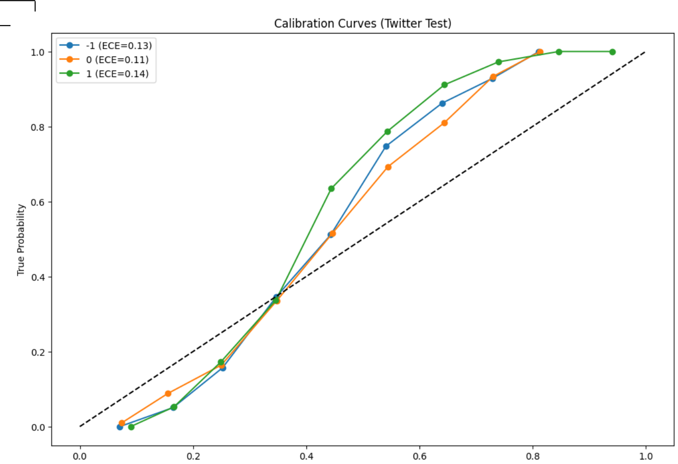

Fig. 5. Calibration curves for Logistic Regression on the Twitter test set. The dashed diagonal represents perfect calibration. All three classes show overconfidence, particularly at high predicted probability values.

Table V presents the Expected Calibration Error (ECE) per class on the Twitter test set.

TABLE V
EXPECTED CALIBRATION ERROR (ECE) PER CLASS ON TWITTER TEST SET

Sentiment Class	ECE
Negative (-1)	0.129
Neutral (0)	0.111
Positive (1)	0.144

The Positive class has the highest ECE (0.144), meaning its probability outputs are least reliable when applied cross-domain. The overconfidence finding is critical: a deployed model would not only produce wrong predictions but also assign high confidence to those incorrect predictions, making them harder to detect and filter in downstream applications.

IX. COMPARISON WITH EXISTING SYSTEMS

A. Performance Comparison

Table VI compares the baseline Logistic Regression model with its CORAL-adapted counterpart on the Twitter test set.

TABLE VI
CORAL DOMAIN ADAPTATION VS. BASELINE LOGISTIC REGRESSION (TWITTER TEST)

Model	Accuracy	Weighted F1
Baseline Logistic Regression	0.6333	0.6332
CORAL-Adapted Logistic Regression	0.5943	0.5865
Improvement	-0.039	-0.047

B. Discussion

CORAL adaptation yielded a *negative* improvement of -0.047 in weighted F1. Several factors explain this outcome:

1) **High-dimensional sparsity:** TF-IDF features are very high-dimensional and sparse. Covariance alignment (using $3,000 \times 3,000$ matrices) is numerically unstable in this regime, even with regularization ($\epsilon = 10^{-3}$).
2) **Semantic rather than statistical mismatch:** The vocabulary mismatch between Reddit and Twitter is a semantic problem—Twitter uses different words to express the same sentiments. CORAL corrects second-order statistics but cannot bridge semantic gaps.
3) **Vocabulary OOV:** Twitter-specific tokens (slang, political handles, hashtag-derived words) are simply absent from the Reddit-trained TF-IDF vocabulary, making feature-level alignment structurally incomplete.

These findings are consistent with [6], which concludes that feature-level domain adaptation is insufficient for text and that neural methods such as fine-tuned BERT [7] are necessary for robust cross-domain NLP.

C. Advantages of the Proposed Analysis

Despite CORAL not improving performance, the analytical contributions of this work—combining linguistic compression analysis with calibration analysis—provide actionable insights:

- Identifies *why* models fail, not merely that they fail.
- Highlights which sentiment classes carry the most risk in cross-platform deployment.
- Establishes a framework for assessing real-world risk before deploying social media NLP models in high-stakes applications.

X. MENTAL HEALTH RISK ASSESSMENT

A. Framing

From a mental health monitoring perspective, failure to correctly detect *Negative* sentiment in user-generated content—posts expressing distress, depression, or suicidal ideation—can have serious real-world consequences. We treat the Negative sentiment class (-1) as the high-risk class and compute the *missed case rate*:

$$\text{Missed Rate} = \frac{\Delta F1_{\text{class}}}{F1_{\text{Reddit, class}}} \tag{1}$$

B. Risk Findings

Table VII presents the complete risk analysis across all three sentiment classes.

TABLE VII
SENTIMENT RISK ANALYSIS: F1 DROP AND MISSED CASE RATE

Class	RD F1	TW F1	Drop	Missed%
Neutral (Low Risk)	0.751	0.672	0.080	10.60%
Positive (Low Risk)	0.730	0.656	0.074	10.11%
Negative **(High Risk)**	0.678	0.631	0.047	**6.96%**

Fig. 6 visualizes the F1 drop per class, with the Negative class highlighted in red.

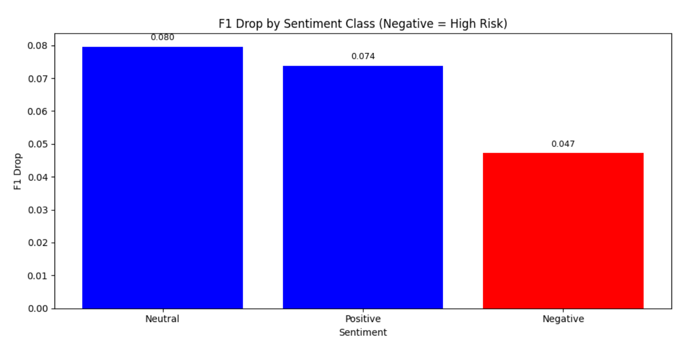

Fig. 6. F1 drop by sentiment class. Negative sentiment (red) is the clinically high-risk class. Despite having the smallest absolute F1 drop, a missed case rate of 6.96% is non-trivial in a mental health monitoring application.

Notably, the Negative class has the smallest absolute F1 drop (0.047) but remains the most clinically significant failure mode. A 6.96% missed case rate in a mental health context means that approximately 1 in 14 distress-expressing posts would be silently ignored by a model deployed without domain adaptation, potentially delaying intervention.

TABLE VIII
ESTIMATED PROJECT BUDGET

Item	Details	Est. Cost (USD)
Cloud Compute	Google Colab Free Tier	\$0
Cloud Compute	Google Colab Pro (optional)	\$10/month
Dataset	Reddit Data (Kaggle, public)	\$0
Dataset	Twitter Data (Kaggle, public)	\$0
API Costs	No external APIs used	\$0
Libraries	All open-source (scikit-learn, NLTK, etc.)	\$0
Hardware	Standard CPU/Colab GPU	\$0
Internet	Standard broadband	∼\$5/month
Total		**\$0 – \$15**

XI. BUDGET ESTIMATION

Table VIII provides the complete project budget.

The entire project was executed within the free tier of Google Colab using only open-source datasets and libraries. All four baseline models, including CORAL adaptation on dense matrices of shape $(2400, 3000)$, ran comfortably within Colab's standard CPU allocation. Optional upgrade to Colab Pro may be required for larger-scale experiments or transformer-based extensions.

XII. ETHICAL CONCERNS

A. Data Privacy

Both datasets are publicly available aggregations with no personally identifiable information (PII). However, the original source data—Reddit comments and tweets—was generated by real individuals. While the datasets are anonymized, their existence raises questions about informed consent in large-scale social media research.

B. Bias and Fairness

Reddit and Twitter datasets reflect demographic, cultural, and linguistic biases inherent to each platform's user base. Reddit skews toward younger, English-speaking, Western users. Twitter is geographically more diverse but carries its own demographic biases. Models trained on such data may perform poorly for users whose language style does not match the training corpus, potentially disadvantaging non-native English speakers or users from underrepresented communities.

C. Mental Health Application Risks

When sentiment analysis models are proposed for mental health applications, overconfident or inaccurate predictions can lead to harmful outcomes: missed detections (false negatives for distress) or false alarms. Our calibration analysis explicitly quantifies this risk: the model is systematically overconfident on cross-domain data, making it unsuitable for deployment without domain-specific re-calibration and validation.

D. Responsible AI Usage

This work does not deploy any system in a real-world application. The analysis is intended to *demonstrate risks* and advocate for domain-aware model design rather than to enable surveillance or monitoring without consent. Any practical deployment of sentiment analysis for mental health applications should involve clinical validation, human-in-the-loop oversight, and explicit user consent.

XIII. CONCLUSION AND FUTURE WORK

A. Summary of Findings

This paper investigated cross-platform sentiment classification by training four classical machine learning models on Reddit and evaluating them on Twitter. The key findings are:

1) All four models exhibit a measurable performance drop when transferred from Reddit to Twitter (ranging from 9.3% to 11.8% accuracy drop), confirming the presence of cross-platform domain shift. Linear SVM achieves the best cross-domain performance with a weighted F1 of 0.653.
2) Reddit posts average 28.6 words versus 20.3 words for Twitter, a structural gap that drives vocabulary mismatch. The TF-IDF feature matrix contains 4,971 effective features from the Reddit vocabulary, many of which have no coverage in Twitter.
3) Linguistic compression analysis reveals a Pearson correlation of $r = 0.12$ between Type-Token Ratio compression and F1 performance drop, suggesting vocabulary richness differences are one (but not the only) driver of generalization failure.
4) Calibration analysis shows that classifiers are overconfident on cross-domain data, with ECE values of 0.13, 0.11, and 0.14 for Negative, Neutral, and Positive classes respectively.
5) CORAL domain adaptation produced a negative improvement of -0.047 in weighted F1, demonstrating that statistical feature-level alignment is insufficient for cross-platform text classification where semantic gaps are the primary challenge.
6) The Negative sentiment class has a missed case rate of 6.96% in cross-domain deployment, posing the highest real-world risk for mental health monitoring applications.

B. Possible Improvements and Future Work

Future directions include:

- **Transformer-based models:** Fine-tuning BERT [7], RoBERTa, or domain-specific models such as BERTweet [8] would substantially reduce domain shift due to contextual representations and pre-training on diverse social media text.
- **Advanced domain adaptation:** Adversarial domain adaptation, domain-invariant feature learning, or cross-lingual transfer could be explored as alternatives to CORAL.
- **Multi-platform training:** Training on a mixture of Reddit and Twitter data, or using meta-learning approaches, may improve cross-platform robustness.
- **Finer-grained emotion labels:** Extending beyond three-class sentiment to fine-grained emotions (joy, anger, fear, sadness) would increase clinical utility.

- **Temporal analysis:** Investigating how sentiment models degrade over time (temporal domain shift) would be valuable for long-term deployed systems.
- **Temperature scaling:** Post-hoc calibration (temperature scaling) could reduce ECE before cross-domain deployment without requiring model retraining.

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